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import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
import seaborn as sns
import matplotlib.pyplot as plt
df = sns.load_dataset('titanic')
print(df.head())
print("\n===== Dataset Information =====")
print(df.info())
print("\n===== Statistical Summary =====")
print(df.describe())
print("\n===== Missing Values =====")
print(df.isnull().sum())
df['age'] = df['age'].fillna(df['age'].median())
df['embark_town'] = df['embark_town'].fillna(df['embark_town'].mode()[0])
df.drop(columns=['deck'], inplace=True)
le = LabelEncoder()
df['sex'] = le.fit_transform(df['sex'])
df.drop(columns=['embarked', 'class', 'who', 'alive', 'adult_male', 'alone'], inplace=True)
scaler = StandardScaler()
numerical_cols = ['age', 'fare']
df[numerical_cols] = scaler.fit_transform(df[numerical_cols])
X = df.drop('survived', axis=1)
y = df['survived']
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
print("\nTraining Data Shape:", X_train.shape)
print("\nTest Data Shape:", X_test.shape)
print("\nFirst 5 Rows of Training Data:")
print(X_train.head())

```

```

0    survived      891 non-null    int64
1    pclass        891 non-null    int64
2    sex           891 non-null    object
3    age           714 non-null    float64
4    sibsp         891 non-null    int64
5    parch         891 non-null    int64
6    fare          891 non-null    float64
7    embarked     889 non-null    object
8    class         891 non-null    category
9    who           891 non-null    object
10   adult_male    891 non-null    bool
11   deck          203 non-null    category
12   embark_town   889 non-null    object
13   alive         891 non-null    object
14   alone         891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
None

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===== Statistical Summary =====

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	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000

```
Training Data Shape: (712, 7)
Test Data Shape: (179, 7)
```

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First 5 Rows of Training Data:
```

	pclass	sex	age	sibsp	parch	fare	embark_town
331	1	1	1.240235	0	0	-0.074583	Southampton
733	2	1	-0.488887	0	0	-0.386671	Southampton
382	3	1	0.202762	0	0	-0.488854	Southampton
704	3	1	-0.258337	1	0	-0.490280	Southampton
813	3	0	-1.795334	1	2	-0.018709	Southampton